

From Integration Bees to Euler-Maclaurin and Beyond: the Floor-Function Integral & Summation via Differential Equations

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This note discusses interesting developments originating from a common class of elementary integrals; those which contain floor-functions. As an example, to solve the following;

[Cambridge Admissions Testing] What is the value of $\int_0^{99} 2^{\lfloor x \rfloor} dx$?

The natural solution would be to break apart the interval into cases before summing $\int_0^1 2^0 + \int_1^2 2^1 + \dots + \int_{98}^{99} 2^{98} = 2^0 + 2^1 + \dots + 2^{98} = 2^{99} - 1$. In doing so, the ‘floor-function-integral’ translates a continuous form of summation into a *discrete* sum, and this interplay between the continuous and the discrete will be at the heart of the work presented here.

1. In *Something Fun*, we develop helpful formulas to tackle floor-function integrals commonly found in Integration Bees,
2. In *Something Old*, these formulas are used to construct and motivate the Euler-Maclaurin Summation Formula (EMS),
3. In *Something New*, findings and worked examples regarding an unusual method of summation are presented,
4. And finally in *Bonus Material*, a notational shorthand for trigonometric simplification is demonstrated, for use in, for example, integration bees where speed is of importance.

1 Something Fun: Floor Function Integrals

We begin by outlining the following pair of formulas for evaluating a family of discrete, floor-function-integrals, and its analogous counterpart in the continuous case; ($m, n \in \mathbb{Z}$)

$$\int_m^n \lfloor x \rfloor g'(x) dx = ng(n) - (m-1)g(m) - \sum_{k=m}^n g(k) \quad (1)$$

$$\int_m^n x g'(x) dx = ng(n) - mg(m) - \int_m^n g(k) dk$$

This intriguing correspondence between the discrete and continuous versions of the integral follows naturally from considering a *triangular* view of the sub-divided interval $[m, n)$. Of course, in each interval $[k, k+1)$, we have $\lfloor x \rfloor = k$:

$$m \int_m^{m+1} g' + (m+1) \int_{m+1}^{m+2} g' + \dots + (m+n-m-1) \int_{n-1}^n g' = \int_m^n \lfloor x \rfloor g'(x) dx$$

By ‘triangular’, I mean to say separating out the integrals in the sum above into rows based on their coefficients:

$$\begin{aligned}
m \int_m^{m+1} g' + m \int_{m+1}^{m+2} g' + m \int_{m+2}^{m+3} g' + \cdots + m \int_{n-1}^n g' &= mg(n) - mg(m) \\
&+ \int_{m+1}^{m+2} g' + \int_{m+2}^{m+3} g' + \cdots + \int_{n-1}^n g' = g(n) - g(m+1) \\
&+ \int_{m+2}^{m+3} g' + \cdots + \int_{n-1}^n g' = g(n) - g(m+2) \\
&\vdots \\
&+ \int_{n-1}^n g' = g(n) - g(n-1)
\end{aligned}$$

Summing and simplifying the RHS thus concludes the proof.

$$m \int_m^{m+1} g' + (m+1) \int_{m+1}^{m+2} g' + \cdots + (m+n-m-1) \int_{n-1}^n g' = (m+n-m-1)g(n) - mg(m) - \sum_{k=m+1}^{n-1} g(k)$$

An equivalent form of this floor-function-integral formula comes from applying the following u -substitution, where we insist that $g^{-1}(a), g^{-1}(b) \in \mathbb{Z}$ and $g^{-1}(b) \geq g^{-1}(a)$;

$$\int_a^b \lfloor g^{-1}(x) \rfloor dx = \int_{g^{-1}(a)}^{g^{-1}(b)} \lfloor u \rfloor g'(u) du = bg^{-1}(b) - a(g^{-1}(a) - 1) - \sum_{k=g^{-1}(a)}^{g^{-1}(b)-1} g(k) \quad (2)$$

Popular Appearances: Integration Bees

Formulas (1) and (2) were inspired by, and are surprisingly useful in, the context of integration bees, as will be demonstrated with a collection of integral-bee problems in order of increasing complexity:

- *Caltech Harvey Mudd Math Contest 2023* Using (2).

$$\int_0^1 \lfloor \log_{2023}(t) \rfloor dt \Rightarrow g(x) = 2023^x \Rightarrow 1 \cdot \log_{2023}(1) - \lim_{a \rightarrow 0} a(\log_{2023}(a) - 1) - \sum_{-\infty \leq k \leq 0} 2023^k = -\frac{2023}{2022}$$

- *MIT 2023 Quarterfinals #4* Using (2).

$$\begin{aligned}
\int_0^1 \left\lfloor \sqrt{1 + \frac{1}{x}} \right\rfloor dx &= \frac{2}{3} - \int_{\frac{1}{3}}^0 \left\lfloor \sqrt{1 + \frac{1}{x}} \right\rfloor dx \Rightarrow g(x) = \frac{1}{x^2 - 1} \\
\frac{2}{3} - \left[\lim_{b \rightarrow 0} b \sqrt{1 + \frac{1}{b}} - \frac{1}{3} \left(\sqrt{1 + \frac{1}{1/3}} - 1 \right) - \sum_{2 \leq k \leq \infty} \frac{1}{k^2 - 1} \right] &= 1 + \frac{1}{2} \sum_{k=2}^{\infty} \frac{1}{k-1} - \frac{1}{k+1} = \frac{7}{4}
\end{aligned}$$

Where, since $\sum_{n_1}^{n_2} \neq -\sum_{n_2}^{n_1}$, one must be careful to maintain the ordering on $g^{-1}(b) \geq g^{-1}(a)$.

- *MIT Finals Tiebreaker 2024* Using (1).

$$\begin{aligned}
\int_2^{\infty} \lfloor x \rfloor \frac{x^2}{x^6 - 1} dx \Rightarrow g'(x) &= \frac{x^2}{x^6 - 1} \Rightarrow g(x) = \int \frac{x^2}{x^6 - 1} dx = \frac{1}{2} \int \frac{x^2}{x^3 - 1} - \frac{x^2}{x^3 + 1} = \frac{1}{6} \ln \left(\frac{x^3 - 1}{x^3 + 1} \right) \\
&\Rightarrow \lim_{n \rightarrow \infty} \frac{n}{6} \ln \left(\frac{n^3 - 1}{n^3 + 1} \right) - \frac{1}{6} \ln \left(\frac{7}{9} \right) - \frac{1}{6} \sum_{k=2}^{\infty} \ln \left(\frac{k^3 - 1}{k^3 + 1} \right)
\end{aligned}$$

Simplifying and factoring;

$$-\frac{1}{6} \ln \left(\frac{7}{9} \right) - \frac{1}{6} \lim_{N \rightarrow \infty} \left[\sum_{k=2}^N \ln \left(\frac{k-1}{k+1} \right) + \sum_{k=2}^N \ln \left(\frac{(2k+1)^2 + 3}{(2k-1)^2 + 3} \right) \right]$$

Giving the telescoping sum;

$$-\frac{1}{6} \ln \left(\frac{7}{9} \right) - \frac{1}{6} \lim_{N \rightarrow \infty} \left[\ln(1) + \ln(2) - \ln(3^2 + 3) + \ln \left(\frac{(2N+1)^2 + 3}{N(N+1)} \right) \right]$$

$$= \frac{1}{6} \ln \left(\frac{27}{14} \right)$$

Towards a more productive application, one can easily deduce an important *integral representation of the ζ function* using (1):

$$I(s) = \int_1^\infty s x^{-s-1} (1 - \{x\}) dx = 1 - \frac{s}{s-1} - \int_1^\infty -s x^{-s-1} \lfloor x \rfloor dx$$

$$\Rightarrow g(x) = x^{-s} \Rightarrow -\frac{1}{s-1} + \lim_{n \rightarrow \infty} n \cdot n^{-s} - (1-1) \cdot 1^{-s} - \sum_{k=1}^n \frac{1}{k^s} = \zeta(s) - \frac{1}{s-1}$$

Where s is a complex variable. Although we have assumed that $\Re(s) > 1$ to evaluate $I(s)$, it is clear that it exists, is analytic, and is bounded $\leq \int_1^\infty s x^{-s-1} dx = 1$ even for $0 < \Re(s) \leq 1$.

And so we arrive at the standard fact that $\zeta(s)$ admits an analytic continuation in the critical strip with a simple pole of residue 1 at $s = 1$.

Exercises

As a great enthusiast of integration bees, I could not help but leave more floor-function-integrals appearances as exercises.

1. (*MIT Int. Bee 2025 Finals*) Using formula (2) and the usual $\{x\} = x - \lfloor x \rfloor$, evaluate:

$$\int_0^{2025} \{\sqrt{x}\} dx$$

2. (*Carnegie Mellon Informatics & Mathematics Competition 2023*) Recalling the Taylor Series for $\ln(1-x) = -\sum_{k=1}^\infty \frac{x^k}{k}$, evaluate:

$$\int_0^\infty e^{-\lfloor x \rfloor (1 + \{x\})} dx$$

3. What do you expect $\int_a^b c^{-1} \lfloor cx \rfloor dx$, $c \in \mathbb{R}$ to be asymptotically equal to as $c \rightarrow \infty$? Now, consider $\int_a^b c^{-1} f(\lfloor cx \rfloor) dx$. Prove that for continuous functions f , this is equal to f 's discrete *Riemann Sum* and that:

$$\lim_{c \rightarrow \infty} \int_a^b c^{-1} f(\lfloor cx \rfloor) dx = \int_a^b f(x) dx$$

2 Something Old: Euler–Maclaurin Summation

Subtracting the discrete and continuous versions of formula (1), one begins a well-trodden path towards approximating the *sum* of a function at integer arguments by its *integral* over that interval;

$$\sum_{k=m}^n f(k) - \int_m^n f(k) dk - \frac{f(n) + f(m)}{2} = \int_m^n \left(x - \lfloor x \rfloor - \frac{1}{2} \right) f'(x) dx = S$$

The *Euler-Maclaurin Summation Formula* (EMS) provides an important and general formula for the approximation of the remainder term S for functions whose derivatives eventually become close to zero.

The importance in understanding the remainder term S cannot be overstated: number-theoretic sums can be approximated by integrals, convergence questions about continuous problems can be equated to discrete ones, and all manner of finite, numerical methods for integration can be rigorously analyzed.

Here, we will *motivate* a derivation of the leading terms of the EMS solely from the definition of S above.

Putting $r(x) = x - \lfloor x \rfloor - \frac{1}{2}$, we notice that r has mean value 0 over its period, and so for functions whose first derivative is roughly constant, $S \approx 0$. Thus, we have found that the leading term of the EMS is;

$$\sum_{k=m}^n f(k) \approx \int_m^n f(k) dk + \frac{f(n) + f(m)}{2}$$

The error in this approximation can be easily bounded by an application of Cauchy-Schwarz;

$$|\varepsilon| = \left| \int_m^n r(x) f'(x) dx \right| \leq \sqrt{n-m} \sqrt{\int_m^n |r(x)|^2 dx} \sqrt{\frac{1}{n-m} \int_m^n |f'(x)|^2 dx} = \frac{(n-m)}{\sqrt{12}} |f'|_{\text{RMSE}}$$

The core mechanism to improve the approximation comes from placing more and more derivatives onto f by the process of integration by parts, as facilitated by a new function $R(x)$ which satisfies $R''(x) = x$.

$$S = \sum_{k=m}^{n-1} \int_k^{k+1} r(x) f'(x) dx = \sum_{k=m}^{n-1} \int_k^{k+1} R''(r(x)) f'(x) dx$$

The derivatives are 'transferred' via IBP;

$$= \sum_{k=m}^{n-1} \left[R' \left(\frac{1}{2} \right) f'(k+1) - R' \left(\frac{-1}{2} \right) f'(k) - R \left(\frac{1}{2} \right) f''(k+1) + R \left(\frac{-1}{2} \right) f''(k) + \int_k^{k+1} f'''(x) R(r(x)) dx \right]$$

As values for f on $m < k < n$ are 'unknown', it is imperative that we set $R(1/2) = R(-1/2)$ and $R'(1/2) = R'(-1/2)$ such that the sum conveniently *telescopes*;

$$S = R' \left(\pm \frac{1}{2} \right) [f'(n) - f'(m)] - R \left(\pm \frac{1}{2} \right) [f''(n) - f''(m)] + \int_m^n f'''(x) R(r(x)) dx$$

The solution to $R''(x) = x$, $R(1/2) = R(-1/2)$, $R'(1/2) = R'(-1/2)$ is of course a cubic polynomial whose coefficients turn out to be $\frac{1}{6}x^3 - \frac{1}{24}x + d$.

And as before, ensuring that $R(r(x))$ has a mean value of zero over $-1/2 \leq r(x) \leq 1/2$ provides an especially accurate, unbiased approximation. Fixing $d = 0$, gives $R(1/2) = 0$, $R'(1/2) = 1/12$ and yields;

$$\sum_{k=m}^n f(k) \approx \int_m^n f(k) dk + \frac{f(m) + f(n)}{2} + \frac{f'(n) - f'(m)}{12}$$

Which provides the first two terms in the EMS expansion. As a point of comparison, our new remainder term has a now smaller magnitude bounded by;

$$|\varepsilon| = \left| \int_m^n R(r(x)) f'''(x) dx \right| \leq \sqrt{n-m} \sqrt{\int_m^n |R(r(x))|^2 dx} \sqrt{\frac{1}{n-m} \int_m^n |f'''(x)|^2 dx} = \frac{(n-m)}{\sqrt{30240}} |f'''|_{\text{RMSE}}$$

Which becomes particularly more accurate if f has decaying derivatives.

The reader is encouraged to think about the nature of continuing this process; how constructing polynomials, R , of higher and higher degree will allow us to transfer more derivatives onto f such that its variation becomes smaller and smaller, all-the-while tuning R so that it's integral is as small as possible.

Exercises

1. Observe that the two-term EMS found previously has an error only on the order of the third derivative of f . Obtain an *exact* formula for the sum of $1 + 2 \cdots + n$.
2. Euler famously calculated the value of $\frac{1}{1^2} + \frac{1}{2^2} + \cdots = \frac{\pi^2}{6}$ to an astounding accuracy using only a small number of terms and the eponymous EMS.

Form your own estimation of $\pi^2/6$ by selecting a value of m after which the infinite series must be estimated by the EMS.

$$\frac{\pi^2}{6} = \sum_{k=1}^{m-1} \frac{1}{k^2} + \sum_{k=m}^{\infty} \frac{1}{k^2}$$

Where n has been taken $\rightarrow \infty$. How accurate will your estimate for π be? Why are decreasing functions like $1/x^2$ especially amenable to EMS estimation?

3. Verify that $R(x) = \frac{1}{120}x^5 - \frac{1}{144}x^3 + \frac{7}{5760}x$ is the next polynomial in the EMS sequence with $R''''(x) = x$, $R^{(j)}(1/2) = R^{(j)}(-1/2)$ $0 \leq j \leq 3$. The resulting RMSE error coefficient is thus added to the sequence $\frac{1}{\sqrt{12}} \approx 0.289$, $\frac{1}{\sqrt{30240}} \approx 0.00575$, $\sqrt{\frac{2701}{7664025600}} \approx 0.000594$

Distributional Connections

In EMS, the core instrument which translates the discrete to the continuous is the function $r(x) = x - \lfloor x \rfloor - 1/2$ and, in a distributional sense,

In the theory of the Fourier Transform, an important object is the periodic Dirac-delta distribution, $\tilde{\delta}(x)$, defined by the frequency-selecting property;

$$\int_{-\infty}^{\infty} \tilde{\delta}(x) f(x) dx = \sum_{k=-\infty}^{\infty} f(k)$$

Having a period of 1, $\tilde{\delta}$ admits a formal Fourier series of the form $\tilde{\delta}(x) = \sum_{k \in \mathbb{Z}} e^{2\pi i k x}$, where $e^{2\pi i k x}$ corresponds to the Fourier component of $\delta(x - k)$.

Connecting this back to $r(x)$, we find a direct relationship between the two periodic distributions $\tilde{\delta}(x) - 1$ and $-r'(x)$;

$$\int_{-\infty}^{\infty} [\tilde{\delta}(x) - 1] f(x) dx = \sum_{k=-\infty}^{\infty} f(k) - \int_{-\infty}^{\infty} f(k) dk = S = \int_{-\infty}^{\infty} r(x) f'(x) dx = - \int_{-\infty}^{\infty} r'(x) f(x) dx$$

Wonderfully, formal manipulation of these series gives rise to the legitimate Fourier expansion for the sawtooth wave;

$$r(x) = \int 1 - \tilde{\delta} = \int 1 - \sum_{k=-\infty}^{\infty} e^{2\pi i k x} dx = \int -2 \sum_{k=1}^{\infty} \cos(2\pi k x) dx = - \sum_{k=1}^{\infty} \frac{\sin(2\pi k x)}{\pi k}$$

Matching perfectly with the sawtooth that is $r(x)$.

Another particularly important appearance of the floor-function integral in the theory of distributions is in the connection between continuous probability distributions p and their discrete counterparts.

By expanding out the following *Riemann–Stieltjes Integral* with respect to a mesh π and differential $d\lfloor x \rfloor$, we have;

$$\int_{\mathbb{R}} f(x) p(x) d\lfloor x \rfloor = \lim_{|\pi| \rightarrow 0} \sum_{x_k \in \pi} f(x_k^*) p(x_k^*) (\lfloor x_{k+1} \rfloor - \lfloor x_k \rfloor)$$

Where $x_k^* \in [x_k, x_{k+1}]$. We see that the only terms which remain in this sum are those x_{k+1}, x_k pairs which straddle an integer;

$$= \lim_{\max(\epsilon) \rightarrow 0} \sum_{n \in \mathbb{Z}} f(n^*) p(n^*) (\lfloor n + \epsilon \rfloor - \lfloor n - \epsilon \rfloor) = \sum_{n \in \mathbb{Z}} f(n) p(n)$$

Which thus links expectation, variance, entropy, etc. of discrete distributions to continuous ones via the $dp = p d\lfloor x \rfloor$ differential.

3 Something New: Summation by Differential Equations

We now turn to a delightful innovation on the premise of approximating a sum via its integral: finding an exact closed-form for the sum of a function via the solution to a boundary-value differential equation.

Unfortunately, this method appears to be restricted to functions that are the solution to a *constant coefficient linear* differential equation $\mathcal{D}[f] = 0$. Our approach will be to construct $R(x)$ from the solution to the following adjoint differential equation;

$$\mathcal{D}^\dagger[R] = x, R(1/2) = R(-1/2), R'(1/2) = R'(-1/2), \dots$$

Where $\mathcal{D}^\dagger[R]$ is defined by the usual transferring of derivatives from integration by parts:

$$\int_k^{k+1} \underbrace{\mathcal{D}^\dagger[R](r(x))}_{:=r(x)} f'(x) dx = [\text{Boundary Terms}] \pm \int_k^{k+1} R(r(x)) \underbrace{\mathcal{D}[f](x)}_{:=0} dx = [\text{Boundary Terms}]$$

This arrangement ensures that the integral remainder in S actually *vanishes*, leaving the boundary terms as the precise result of the summation. As per usual, constraints on R at $\pm 1/2$ are in place to ensure telescoping when summed $m \leq k \leq n-1$.

Worked Example

As a concrete example, suppose we wished to find the closed-form sum of sinusoidal functions f s.t. $f''(x) + \omega^2 f(x) = 0$. The weaker statement $f'''(x) + \omega^2 f'(x) = 0$ will be used for reasons soon to be seen.

We would begin by ‘transferring’ derivatives from $\mathcal{D}$ onto R until only f' remains in the integrand;

$$\begin{aligned} 0 &= \int_k^{k+1} R(r(x)) \underbrace{[f'''(x) + \omega^2 f'(x)]}_{\mathcal{D}[f']} dx \\ &= \left[R\left(\frac{1}{2}\right) f''(k+1) - R\left(\frac{-1}{2}\right) f''(k) \right] - \left[R'\left(\frac{1}{2}\right) f'(k+1) - R'\left(\frac{-1}{2}\right) f'(k) \right] + \int_k^{k+1} \underbrace{[R''(r(x)) + \omega^2 R(r(x))]}_{\mathcal{D}^\dagger[R]} f'(x) dx \end{aligned}$$

As the operator $\mathcal{D}$ is linear and constant-coefficient, this regrouping will always be possible and will result in a new linear constant-coefficient differential operator on R with alternating signs.

Thus, determining R s.t. $R(1/2) = R(-1/2), R'(1/2) = R'(-1/2)$, and $R''(x) + \omega^2 R(x) = x$ would immediately imply that;

$$-S = \sum_{k=m}^{n-1} - \int_k^{k+1} r(x) f'(x) dx = R\left(\frac{1}{2}\right) \sum_{k=m}^{n-1} [f''(k+1) - f''(k)] - R'\left(\frac{1}{2}\right) \sum_{k=m}^{n-1} [f'(k+1) - f'(k)]$$

Giving a delightful, telescoping solution to the original sum over f ;

$$\sum_{k=m}^n f(k) = \int_m^n f(k) dk + \frac{f(n) + f(m)}{2} - R(1/2)[f''(n) - f''(m)] + R'(1/2)[f'(n) - f'(m)]$$

A quick application of undetermined coefficients yields the desired solution for R ;

$$x = \mathcal{D}^\dagger[R] = R'' + \omega^2 R \Rightarrow R(x) = \frac{x}{\omega^2} - \frac{1}{2\omega^2} \frac{\sin(\omega x)}{\sin(\frac{\omega}{2})}$$

And we are done! With this kernel in hand, any combination of $f(x) = \cos(\omega x)$ and $f(x) = \sin(\omega x)$ can be summed exactly, for any value of m, n , without any additional work.

One can effortlessly recover generalized versions of Lagrange's Trigonometric Identity;

$$\sum_{k=m}^n \cos(\omega k) = \cot\left(\frac{\omega}{2}\right) \frac{\sin(\omega n) - \sin(\omega m)}{2} + \frac{\cos(\omega n) + \cos(\omega m)}{2}$$

And its sibling;

$$\sum_{k=m}^n \sin(\omega k) = -\cot\left(\frac{\omega}{2}\right) \frac{\cos(\omega n) - \cos(\omega m)}{2} + \frac{\sin(\omega n) + \sin(\omega m)}{2}$$

Without any kind of tedious appeal to complex numbers or induction.

Perhaps the most beautiful property of the kernel $R(x)$, is that as the frequency of the oscillators tend to zero, a polynomial kernel is produced;

$$\lim_{\omega \rightarrow 0} \frac{x}{\omega^2} - \frac{1}{2\omega^2} \frac{\sin(\omega x)}{\sin(\frac{\omega}{2})} = \frac{x^3}{6}$$

And so from the ashes of sinusoids rises the triangular numbers: since $f(x) = x$ satisfies $f'' + 0^2 f = 0$, we may employ summation by differential equations to conclude that;

$$\sum_{k=m}^n k = \int_m^n k dk + \frac{n+m}{2} - R(1/2)[0] + R'(1/2)[0] = \frac{n^2 - m^2 + n + m}{2}$$

General Linear Case

Since the solutions to constant-coefficient linear ODEs are, after all, simply linear combinations of exponentials, their finite sum poses no more challenge than a simple geometric series.

$$\sum_{k=m}^n \sum_i c_i e^{\lambda_i x} = \sum_i c_i \frac{e^{\lambda_i(n+1)} - e^{\lambda_i m}}{e^{\lambda_i} - 1}$$

In any case, there is an undeniable elegance to the method of summation by differential equations which cannot help but be explored further.

Using the fact that a general linear system $\frac{d}{dx}\mathbf{y} = \mathbf{A}\mathbf{y}$, $\mathbf{A} \in \mathbb{C}^{p \times p}$ can always be converted into a p -th order constant-coefficient ODE, we will suppose that, $\sum_{l=0}^p c_l f^{(l)}(x) = 0$ and substitute this into our initial template for S ;

$$0 = \int_k^{k+1} R(r(x)) \underbrace{\left[\sum_{l=0}^p c_l f^{(l+1)}(x) \right]}_{\mathcal{D}[f']} dx$$

Integration by parts and recollection of terms produces;

$$= \sum_{l=0}^p c_l \sum_{j=0}^{l-1} (-1)^j \left[R^{(j)}\left(\frac{1}{2}\right) f^{(l-j)}(k+1) - R^{(j)}\left(-\frac{1}{2}\right) f^{(l-j)}(k) \right] + \int_k^{k+1} \underbrace{\left[\sum_{l=0}^p c_l (-1)^l R^{(l)}(r(x)) \right]}_{\mathcal{D}^\dagger[R]} f'(x) dx$$

Unsurprisingly $\mathcal{D}^\dagger$ is simply $\mathcal{D}$ with alternating signs. As before, constraining $R(x)$ to be the solution to the following linear BVP will eliminate the integral term;

$$\sum_{l=0}^p c_l (-1)^l R^{(l)}(x) = x, \quad R^{(j)}(1/2) = R^{(j)}(-1/2) = \rho_j \quad \forall j: 0 \leq j \leq p-1$$

Although it is readily seen that $R(x) = \alpha f(-x) + x/c_0 + c_1/c_0^2$ solves the differential equation, this in general does not match the boundary constraints. However, R will indeed share the same exponentials as $f(-x)$, though with a different set of coefficients.

After summing $m \leq k \leq n$, we are thus left with the final, exact expression;

$$-S = \sum_{l=1}^p c_l \sum_{j=0}^{l-1} (-1)^j \rho_j \left[f^{(l-j)}(n) - f^{(l-j)}(m) \right]$$

For example, the classic geometric series given by $f(x) = r^x$ would have $f'(x) - \ln(r)f(x) = 0$, $c_0 = -\ln(r)$, $c_1 = 1$, $p = 1$, and $-R'(x) - \ln(r)R(x) = x$.

This yields $R(x) = \frac{r^{-x}}{\ln(r)(\frac{1}{\sqrt{r}} - \sqrt{r})} - \frac{x}{\ln(r)} + \frac{1}{\ln(r)^2}$ with $\rho_0 = \frac{1}{\ln(r)} \left(\frac{1}{1-r} - \frac{1}{2} + \frac{1}{\ln(r)} \right)$ giving finally;

$$\sum_{k=m}^n r^k = -c_1 (-1)^0 \rho_0 (\ln(r) r^n - \ln(r) r^m) + \int_m^n r^k dk + \frac{r^n + r^m}{2} = \frac{r^{n+1} - r^m}{(r-1)}$$

As was expected. As in the case of the sinusoids, the efficiency and generality of the method is exchanged for the preparatory difficulty of establishing the kernel R .

In addition, like with the sinusoid example, the *limiting behavior* of this differential kernel about an otherwise undefined point carries deeper structure;

$$\lim_{r \rightarrow 1} \frac{r^{-x}}{\ln(r) \left(\frac{1}{\sqrt{r}} - \sqrt{r} \right)} - \frac{x}{\ln(r)} + \frac{1}{\ln(r)^2} = -\frac{x^2}{2}$$

Which is to say that $\lim_{r \rightarrow 1} \sum_{k=m}^n r^k$ has a well-defined, *polynomial* behavior in m, n (clearly $n-m+1$).

The reader is also invited to consider different ways of simplifying the above expression for S given the starting condition $\mathcal{D}[f] = 0$. For example, since $\int_m^n [c_p f^{(p)}(x) + c_{p-1} f^{(p-1)}(x) \cdots + c_1 f'(x) + f(x)] dx = 0$, we have;

$$-\int_m^n f(x) dx = c_p \left[f^{(p-1)}(n) - f^{(p-1)}(m) \right] + c_{p-1} \left[f^{(p-2)}(n) - f^{(p-2)}(m) \right] \cdots + c_1 [f(n) - f(m)]$$

Which is the $j = 1$ term. Or, for the $j = 0$ term, we see that;

$$\sum_{l=1}^p c_l \left[f^{(l)}(n) - f^{(l)}(m) \right] = -f(n) + f(m)$$

4 Bonus Integration Bee Material: Notational Shortcut for Product-To-Sum

Given a product of cosine functions, $C(x) = \prod_i \cos(\omega_i x)$, it can be inductively shown by the following product-to-sum identity that C is equivalent to a *sum* of cosines; $C(x) = \sum_i \cos(\alpha_i x)$.

$$\cos(\alpha) \cos(\beta) = \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2}$$

I wish to demonstrate that by using a certain ‘shorthand’ for C , ie $C(x) \Leftrightarrow [\omega_1, \omega_2, \dots, \omega_n]$, the action of simplifying such a product into a sum of cosines can be significantly expedited.

Under the list notation, the action of the cosine-product to sum identity becomes;

$$x_n \cdot [x_1, x_2, \dots, x_{n-1}]_{/2^k} = [x_1 + x_n, x_2 + x_n, \dots, x_{n-1} + x_n, x_1 - x_n, x_2 - x_n, \dots, x_{n-1} - x_n]_{/2^{k+1}}$$

Which is to say “duplicate the list then add to one and subtract from the other, before dividing all of the coefficients by 2”.

Example 1: To walk through the usage of this shorthand, let us rapidly derive the expansion of $\cos(x)^3$;

$$\cos(x)^3 \Leftrightarrow 1 \cdot (1 \cdot [1]_{/2^0}) = 1 \cdot [1 + 1, 1 - 1]_{/2^1} = 1 \cdot [2, 0]_{/2^1} = [2 + 1, 0 + 1, 2 - 1, 0 - 1]_{/2^2} = [3, 1, 1, -1]_{/2^2}$$

Which reads as $\cos(x)^3 = \frac{1}{2^2} (\cos(3x) + \cos(x) + \cos(x) + \cos(-x))$, which is of course $4 \cos(x)^3 = \cos(3x) + 3 \cos(x)$.

Example 2: Another delightful showcase of the speed of the cosine-list notation is in tackling the following integral from the *MIT 2024 Integration Bee*;

$$\int [\cos(3x) \cos(5x) \cos(6x) \cos(7x) - \cos(x) \cos(2x) \cos(4x) \cos(8x)] dx$$

Jumping straight to the shorthand;

$$3 \cdot 5 \cdot 6 \cdot [7]_{/2^0} = 3 \cdot 5 \cdot [13, 1]_{/2^1} = 3 \cdot [18, 6, 8, -4]_{/2^2} = [21, 9, 11, -1, 15, 3, 5, -7]_{/2^3}$$

$$1 \cdot 2 \cdot 4 \cdot [8]_{/2^0} = 1 \cdot 2 \cdot [12, 4]_{/2^1} = 1 \cdot [14, 6, 10, 2]_{/2^2} = [15, 7, 11, 3, 13, 5, 9, 1]_{/2^3}$$

We can then recognize that the subtraction of these two cosine-sums cancels all but two terms;

$$\int [21]_{/2^3} - [13]_{/2^3} dx = \frac{1}{2^3} \left(\int \cos(21x) dx - \int \cos(13x) dx \right) = \frac{1}{8} \left(\frac{\sin(21x)}{21} - \frac{\sin(13x)}{13} \right) + C$$

Which yields the desired simplification with only a small and compact amount of work.

Example 3: Finally, we will explore a an intriguing cosine-shorthand proof of the classic infinite product identity that is $\prod_{i=1}^{\infty} \cos(x/2^i) = \frac{\sin(x)}{x}$.

We notice that the set of all sums and differences of these powers of two seem to generate a list of binary expansions;

$$\lim_{N \rightarrow \infty} \cos(x/2) \cos(x/4) \cos(x/8) \dots \cos(x/2^N) \Leftrightarrow \lim_{N \rightarrow \infty} [1/2] \cdot [1/4] \cdot [1/8] \dots [1/2^N] = \lim_{N \rightarrow \infty} \left[\frac{1}{2} \pm \frac{1}{4} \pm \frac{1}{8} \dots \pm \frac{1}{2^N} \right]_{/2^{N-1}}$$

Where the brackets contain all 2^{N-1} possible combinations of +’s and -’s. Each such term thus represents the 2^{N-1} -bit binary expansion of a decimal between 0 and 1, giving a partition which grows increasingly tighter as $N \rightarrow \infty$. Converting out of the shorthand;

$$\Leftrightarrow \lim_{N \rightarrow \infty} \frac{1}{2^{N-1}} \sum_{b=1/2^{N-1}, b \in \text{bin}(N-1)}^{1-1/2^{N-1}} \cos(bx) = \lim_{\Delta b \rightarrow 0} \Delta b \sum_{0 < b_k < 1} \cos(b_k x) = \int_0^1 \cos(bx) db = \frac{\sin(x) - \sin(0)}{x}$$

As was to be shown.

4.1 Mixed Sine & Cosine

A product of purely sine functions, or indeed any mixed combination of sines and cosines, can be stored similarly in the list shorthand, but now one must keep track of many more +/− signs, and of

whether a given list represents sines, S , or cosines, C ;

$$[x_1^\pm, x_2^\pm, \dots, x_n^\pm]^C + [y_1^\pm, y_2^\pm, \dots, y_n^\pm]^S \Leftrightarrow \sum_{k=1}^n \pm \cos(x_k) \pm \sin(y_k)$$

Because the sine function is odd, $[(-\omega)^-]^S \Leftrightarrow +\sin(\omega x)$, while since cosine is even, $[(-\omega)^-]^C \Leftrightarrow -\cos(\omega x)$.

Example 4: To illustrate usage of the shorthand under these complications, we will use the example of $\cos(2x)\sin(3x)\cos(4x)\sin(5x) \Leftrightarrow [2]^C \cdot [3]^S \cdot [4]^C \cdot [5]^S$.

First by using the $\sin(\alpha)\cos(\beta) = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}$ identity, we have;

$$[2]^C \cdot [3]^S \cdot [4]^C \cdot [5]^S = [3+2, 3-2]_{/2}^S \cdot [4]^C \cdot [5]^S$$

And once more;

$$[5, 1]_{/2}^S \cdot [4]^C \cdot [5]^S = [5+4, 1+4, 5-4, 1-4]_{/4}^S \cdot [5]^S$$

Then using $\sin(\alpha)\sin(\beta) = \frac{\cos(\alpha-\beta)-\cos(\alpha+\beta)}{2}$, this becomes;

$$\begin{aligned} [9, 5, 1, 3^-]_{/4}^S \cdot [5]^S &= \left[9-5, 5-5, 1-5, (3-5)^-, (9+5)^-, (5+5)^-, (1+5)^-, (3+5)^+ \right]_{/8}^C \\ &= \left[4, 0, -4, (-2)^-, (14)^-, (10)^-, (6)^-, (8)^+ \right]_{/8}^C \\ &\Leftrightarrow \frac{1}{8} (\cos(4x) + \cos(0x) + \cos(-4x) - \cos(-2x) - \cos(14x) - \cos(10x) - \cos(6x) + \cos(8x)) \end{aligned}$$

Example 5: Or as an additional practice problem, we examine $\sin(x)^3$;

$$\begin{aligned} [1]^S \cdot [1]^S \cdot [1]^S &= \left[1-1, (1+1)^- \right]_{/2}^C \cdot [1]^S = [0, 2^-]_{/2}^C \cdot [1]^S = \left[0+1, (1+2)^-, 1-0, (1-2)^- \right]_{/4}^S = \left[1, 3^-, 1, (-1)^- \right]_{/4}^S \\ &\Leftrightarrow \frac{1}{4} (\sin(x) - \sin(3x) + \sin(x) + \sin(x)) \end{aligned}$$

Leaving $\sin(3x) = 3\sin(x) - 4\sin(x)^3$ as desired.